

Hierarchical Motif Family Chunking Report

Hierarchical clustering of motif families shows transformer weights are organized hierarchically.

Families | Entropy | MSE | CosSim

64 | 1.411 | 0.000139 | 0.220

32 | 1.011 | 0.000211 | 0.613

16 | 0.923 | 0.000260 | 0.461

8 | 0.865 | 0.000295 | 0.330

BPE on family stream: 87.04% length reduction with 100 merges.

FamilyGPT training loss decreasing: 4.402 -> 0.970 in 200 steps.

Layer family analysis shows Family 25 dominates across Attention Q/K/V, MLP and Embeddings.

Conclusion: 16-64 families maintain low MSE and high compressibility, supporting hierarchical motif o